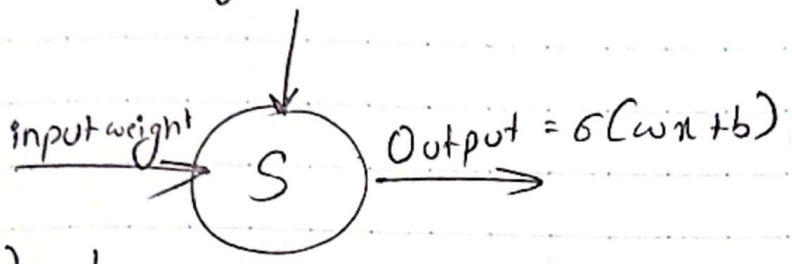
Solve with Sigmoid Neuron



x_1	x_2	x_1 XOR x_2
0	0	0
0	1	1
1	0	1
1	1	0

$\sigma(x) = \frac{1}{1+e^{-x}}$

If $x_1 = 0, x_2 = 0$

$h_1(wx+b) = -0.5$

sigmoid $h_1 = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-(-0.5)}} = \frac{1}{1+e^{0.5}} = 1.6487$

$\therefore h_1 = \frac{1}{1+1.6487} = \frac{1}{2.6487} = 0.377$

$h_2(wx+b) = -0.5$

$= \frac{1}{1+e^{0.6}} = 0.377$

$z_0, ws = 1, w_0 = 1, bias = -0.5$

$= (h_1 \times 1) + (h_2 \times 1) - 0.5$

$= (0.377 \times 1) + (0.377 \times 1) - 0.5$

$= 0.754 - 0.5$

$= 0.254$

Output layer $(o) = \frac{1}{1+e^{-0.254}}$

$= \frac{1}{0.7757}$

$= 0.563$

$x_1 = 0, x_2 = 1 \quad (0,1)$

$zh_1(wx+b) = 0 - 1.5$

$h_1 = \frac{1}{1+e^{-(-1.5)}} = \frac{1}{1+e^{1.5}}$

$= e^{1.5}, 4.4817$

$\therefore h_1 = \frac{1}{1+4.4817} = \frac{1}{5.4817} = 0.182$

$h_2(wx+b) = 0.5$

$h_2 = \frac{1}{1+e^{-(0.5)}} = \frac{1}{1+e^{-0.5}}$

$= e^{-0.5} = 0.6065$

$\therefore h_2 = \frac{1}{1+0.6065} = 0.622$

$z0 = (h_1 \times 1) + (h_2 \times 1) - 0.5$

$= (0.182 \times 2)(0.622 \times 1) - 0.5$

$= 0.804 - 0.5$

$= 0.304$

$(0) = \frac{1}{1+e^{-0.304}}$

$= e^{-0.304}$

$= 0.7379$

$\therefore (0) = \frac{1}{1+0.7379} = 0.575$

$x_1 = 1, x_2 = 0$

$h_1(wx+b) = 0.5$

$h_1 = \frac{1}{1+e^{-0.5}}$

$e^{-0.5} = 0.6065$

$\therefore \frac{1}{1+0.6065} = 0.622$

$h_2(wx+b) = -1.5$

$h_2 = \frac{1}{1+e^{-(-1.5)}} = \frac{1}{1+e^{1.5}}$

$e^{1.5} = 4.4817$

$\therefore h_2 = \frac{1}{1+4.4817} = 0.182$

$z_0 = (h_1 \times 2) + (h_2 \times 1) - 0.5$

$= (0.622 \times 1) + (0.182 \times 1) - 0.5$

$= 0.804 - 0.5$

$= 0.304$

$(0) = \frac{1}{1+e^{-0.304}}$

$= e^{-0.304} = 0.7378$

$= \frac{1}{1+0.7378} = 0.575$

$x_1 = 1, x_2 = 1$

$h_1(wx+b) = -0.5$

$h_1 = \frac{1}{1+e^{-(0.5)}} = \frac{1}{1+e^{+0.5}}$

$= 1.6487$

$\therefore h_1 = \frac{1}{1+1.6487} = 0.377$

$h_2(wx+b) = -0.5$

$h_2 = \frac{1}{1+e^{-(-0.5)}} = \frac{1}{1+e^{0.5}}$

$h_2 = 0.377$

$z_0 = (h_1 \times 1) + (h_2 \times 1) - 0.5$

$= (0.377 \cdot 1) + (0.377 \cdot 1) - 0.5$

$= 0.754 - 0.5$

$= 0.254$

$(0) = \frac{1}{1+e^{-0.254}}$

$= e^{-0.254}$

$= 0.7757$

$\frac{1}{1+0.7757} = 0.563$